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Feng Wu^{a,b}, Yi Xing^a, Li Li^{a,b}, Ji Qian^a, Wenjie Qu^a, Jianguo Wen,^c Dean Miller,^c

Yusheng Ye^a, Renjie Chen^{a,b*}, Khalil Amine,^d Jun Lu^{d*},

a. Beijing Key Laboratory of Environmental Science and Engineering, School of Material Science & Engineering, Beijing Institute of Technology, Beijing 100081, PR China

b. Collaborative Innovation Center of Electric Vehicles in Beijing, Beijing 100081, PR China

c. Electron Microscopy Center, Material Science Division, Argonne National Laboratory, 9700 S. Cass Avenue, Lemont, IL 60439, USA

d. Chemical Science and Engineering Division, Argonne National Laboratory, 9700 S. Cass Avenue, Lemont, Illinois 60439, United States

E-mail address: chenrj@bit.edu.cn (R. Chen); junlu@anl.gov (J. Lu)

KEYWORDS Boron-doped reduction graphite oxide, Catalytic substrate, Discharge capacity, Rate capability, Li-O₂ batteries.

ABSTRACT

To improve the electrochemical performance of the high energy Li-O₂ batteries, it is important to design and construct a suitable and effective oxygen-breathing cathode. Herein, a three-dimensional (3D) porous boron-doped reduction graphite oxide (B-rGO) material with a hierarchical structure has been prepared by a facile freeze-drying method. In this design, boric acid as the boron source helps to form the 3D porous structure, owing to its cross-linking and pore-forming function. This architecture facilitates the rapid oxygen diffusion and electrolyte penetration in the electrode. Meanwhile, the boron-oxygen functional groups linking to the carbon surface or edge serve as additional reaction sites to activate the ORR process. It is vital that boron atoms have been doped into the carbon lattices to greatly activate the electrons in the carbon π system, which is beneficial for fast charge under large current densities. Density functional theory calculation demonstrates that B-rGO exhibits much stronger interactions with Li₅O₆ clusters, so that B-rGO more effectively activates Li-O bonds to decompose Li₂O₂ during charge than rGO does. With B-rGO as a catalytic substrate, the Li-O₂ battery achieves a high discharge capacity and excellent rate capability. Moreover, catalysts could be added into the B-rGO substrate to further lower the overpotential and enhance the cycling performance in future.

1. Introduction

The aprotic Li-O₂ batteries still attract worldwide interests and are considered as one of the promising candidates for electrochemical energy storage devices, due to their extremely high theoretical energy density, which are at least 2–3 times higher than traditional lithium-ion batteries. However, some critical challenges and major problems need to be addressed and overcome before the aprotic Li-O₂ system can be commercialized, such as its low round-trip efficiency, low rate capability, and limited cycle life. The typical aprotic Li-O₂ batteries are usually composed of oxygen-breathing electrodes as cathodes, lithium electrodes as anodes, and aprotic solvents that dissolve lithium salts as electrolytes. It is noteworthy that an ideal cathode can significantly improve the overall system performance, which contains a porous substrate and, in some cases, an added electrocatalyst.¹⁻⁵

Before constructing an effective cathode for Li-O₂ batteries, a suitable porous substrate should first be designed, which could provide numerous tri-phase regions for reactions, enough space for discharge products growth and, in some cases, active sites for catalysts deposition.⁶⁻¹⁰ So far, many kinds of carbon supporting materials have been developed, which is attributed to their low density, low cost, high electronic conductivity and easy synthesis of porous structures, including super P,¹¹ CNTs,¹²⁻¹⁴ reduction graphite oxides (rGO),¹⁵⁻¹⁷ heteroatom-doped graphite carbons,¹⁸⁻¹⁹ etc. It is worth noting that a series of graphite carbon materials doped by heteroatoms (such as N, P, S) have been applied in Li-O₂ batteries as supporting materials or metal-free catalysts, and achieved an improvement in electrochemical performance to some

degree, resulting from promoted conductivity and modified catalytic capability.²⁰⁻²⁴ However, the heteroatom-doped graphite carbons mentioned above all exert a positive influence on the oxygen reduction reaction (ORR), not the oxygen evolution reduction (OER), the serious polarization of which needs to be solved urgently. During OER, the discharge products need to be decomposed, for this reason an ideal substrate or catalyst should be able to help gain electrons from Li_2O_2 to facilitate the decomposition. Boron-doped graphite carbon may be a good candidate owing to its p-type behavior. Zhang's group²⁵ reported the boron-doped graphene reduced the rate-determining step barrier as a catalytic substrate for Li-O₂ batteries determined by first-principles calculation. This may help increase the oxygen evolution rate and improve the rate capability of OER. In the meantime, Wen's group²⁶ also found the boron-doped graphene had good catalytic activity in decreasing the oxygen evolution barrier demonstrated by the first-principles thermodynamic calculations. However, there is no experimental evidence to testify the effect of boron-doped graphite carbon as supporting matrix in aprotic Li-O₂ batteries.

In this work, a porous boron-doped reduction graphite oxide (B-rGO) material was prepared by a facile synthetic approach of freeze-drying, and was applied in aprotic Li-O₂ batteries as the cathode. We chose rGO as the building block to construct the porous heteroatom-doped matrix, owing to its excellent electronic conductivity, high specific surface area and good mechanical strength. In this design, boric acid as the boron source help to form the three-dimensional (3D) porous structure with hierarchical pores during calcination, owing to its cross-linking function of

interconnecting the rGO to each other. Meanwhile, the boron-oxygen groups linking to the carbon surface or edge could serve as additional reaction sites for ORR. It is vital that boron atoms have been doped into the carbon lattices to greatly activate the electrons in the carbon π system, which is beneficial for fast charge under large current densities. The porous B-rGO material as cathode exhibits a high specific capacity and excellent rate capability. To the best of our knowledge, there has only been theoretical research on the role of boron-doped graphite carbon in improving electrochemical performance of Li-O₂ batteries, but no experiment to validate the theories so far.

2. Experimental section

Material preparation

50 mg graphite oxide (GO) was dispersed in 10 ml deionized (DI) water under sonication for 6 h. Then 800 mg boric acid was added into the GO solution, and the mixture was stirred at 35 °C for 2 h. Then the boric acid-GO solution was suddenly immersed into liquid nitrogen to be frozen quickly. The solid-state brown boric acid-GO was dried for 48 h under vacuum in a freezing dryer. The brown boric acid-GO powders were calcined under argon. The temperature was gradually raised to 800 °C at a rate of 5 °C/min, and then kept for 3 h in a tube furnace. Finally, the boron-doped rGO was washed with DI water to further remove the residual boric acid, and was dried at 60 °C under vacuum. The rGO was prepared in the same way without adding boric acid into the GO solution.

Characterization measurement

The N₂ physisorption analysis was carried out at 77 K with a surface characterization analyzer (Nova 2200e Micromeritics Co., USA). The specific surface area was calculated by the Brunauer-Emmett-Teller (BET) method, and the pore-size distribution was obtained from the adsorption branch of the isotherm by using the Barrett-Joyner-Halenda method. The X-ray photoelectron spectroscopy (XPS) experiment was performed with a PHI5300 XPS spectrometer, using Al K α as X-ray radial source, and using C1s spectrum (284.6 eV) to rectify the electronic energy of the elements. The electron paramagnetic resonance (EPR) measurements were performed using a Bruker EMX spectrometer in a cw mode X-band microwave ($f \approx 9.4$ GHz) at room temperature. The field-emission scanning electron microscopy (SEM, Hitachi S-4700) was employed to determine the morphology and estimate the size of the material and discharge products. The cathode laminates after discharge were characterized by high-energy X-ray diffraction (HE-XRD) to determine the formation of Li₂O₂, which was carried out at the 11-ID-D beamline of the Advanced Photon Source, Argonne National Laboratory. The X-ray wavelength was 0.6705 Å. The samples were completely covered by a Kapton tape as a protective film in the glove box to avoid any side reactions from the air. Raman spectra were measured by using a Renishaw 2000 or inVia microscope spectrometer with a HeNe laser at 633 nm wavelength. The sample was loaded inside a glove box into a gas-tight Raman cell with a glass window. Fourier transform infrared (FTIR) spectra were recorded using a Nicolet 6700 FTIR spectrometer. A field-emission transmission electron microscope (TEM; JEOL JEM-2100F FEG FasTEM) with a spherical and chromatic aberration

imaging corrector working at 80 kV was employed to evaluate the samples. The electron energy loss spectroscopy (EELS) measurement was conducted in TEM diffraction mode. The beam was converged to a ~ 2 nm diameter area to enhance the spatial resolution. To prepare the TEM specimens, holey carbon grids were placed in the bottle of as-received powders and were shaken to allow small powders to attach onto the holey carbon.

Electrochemical measurements

Electrochemical characterization was carried out using a Swagelok-type cell, which was composed of a lithium metal anode, glass fiber separator (Fisher Scientific), electrolyte (1M lithium bis(trifluoromethanesulfonyl)imide (LiTFSI) in tetraethylene glycol dimethyl ether (TEGDME)), and a porous cathode (11 mm diameter). The cathode was prepared by casting the sample material and PVDF binder in a mass ratio of 8:2 on carbon paper. The carbon loading density was about $0.3 \text{ mg} \cdot \text{cm}^{-2}$.

The cells were sealed except for the Al grid window that exposed the porous cathode to 1 bar O_2 pressure. The electrochemical measurements were carried out using a MACCOR cycler. The cell was maintained in 1 bar O_2 atmosphere to avoid any negative effects of humidity and CO_2 . We normalized the observed capacity by the total weight of the active material on the carbon paper for comparison in this study.

Theoretical calculation

To obtain the binding energy (BE) of the Li_5O_6 adsorbed on the models, we took the energy difference between the energy of the absorbed system and the energies of the

isolated Li_5O_6 and B-rGO (or rGO). Namely, the BE was obtained as follows: $\text{BE} = E(\text{B-rGO or rGO}) + E(\text{Li}_5\text{O}_6) - E(\text{total})$. Geometry optimizations were carried out with the all-electron density functional program DMol3 in Materials Studio 5.5 (Accelrys) using the Becke–Lee–Yang–Parr (BLYP) functional and the double numerical plus polarization (DNP) basis set. Optimization was considered converged when the following convergence criteria were met: 1.0×10^{-5} Ha for the total energy, 0.002 Ha/Å for the maximum force on atoms, and 0.005 Å for the maximum atomic displacement.

3. Results and discussion

The porous B-rGO material was prepared by a facile freeze-drying process, followed by annealing in inert atmosphere as shown in **Scheme 1**, and the rGO material was also prepared for comparison in the same way. The detail information could be found in the experimental section. In this design, there are two points crucial for the fabrication of the porous B-rGO framework with hierarchical pores. The first one is the boric acid-GO solution should be frozen quickly to avoid the boric acid being dissolved out during cooling. And the frozen powders should be dried by the freeze-drying to develop the 3D macropores structure. The other one is the thermal treatment that breaks down the boric acid to boric oxide and water. Since the boric acid dissolved in the GO solution is massive, the released water vapor and boric oxide gas generates plenty of mesopores, leading to the formation of hierarchical porous architecture. Under a high temperature, the boron atoms are doped into the carbon lattices, while part of boric oxides react with the defects on the surface or edge of the graphene nanosheets. That is, beside being the boron source, the boric oxides are also

regarded as cross-linking agents to interconnect graphene nanosheets together to form the 3D framework. In the meantime, the GO is reduced to graphene.

The structure and morphology of rGO and B-rGO were observed by SEM. As shown in **Figures 1a and 1b**, the rGO is composed of large, thin, wrinkled two-dimensional (2D) graphene sheets with a length of $\sim 10\ \mu\text{m}$, loosely stacked together. In contrast, B-rGO shows a porous hierarchical framework consisting of ultrathin, tablet-like graphene sheets ($\sim 2\ \mu\text{m}$). It is worthy of note that the pores are interconnected inside and outside. The specific surface area and pore size distribution of rGO and B-rGO were determined by BET test. **Fig 1d and 1e** exhibit the pores of B-rGO ranges from meso-size to macro-size, but mainly centered at $\sim 4\ \text{nm}$ (meso-scale), while the pore size of rGO is mostly distributed on the macro-scale. It is already known that the meso-scale pores as active regions are favorable for reactions. The macro-pores are easily flooded by electrolytes, but they could accommodate the growth of Li_2O_2 . The specific surface area of B-rGO is around $200\ \text{m}^2/\text{g}$, and is far larger than that of rGO, as seen from **Figure 1c**. Namely, B-rGO could provide more space for reactions than rGO could. Through discussion above, compared with rGO, B-rGO is more suitable as substrate in Li-O₂ batteries in terms of the material architecture. The TEM images (**Figure 2a and 2b**) and corresponding selected area electron diffraction (SAED) spectrum (**Figure 2c**) confirm that the B-rGO is composed of few graphene layers.

The chemical composition and binding state of B-rGO was investigated by XPS, EELS, ICP and EPR. Through the full range XPS analysis in **Figure 1f**, the boron

peak near 192 eV is only found in the spectrum of B-rGO, which confirms the boron atoms are incorporated into rGO matrix via the high temperature thermal treatment. The content of boron in B-rGO is 1.7 wt%, as determined precisely by ICP. The O content increases from 4.95% to 6.78% for rGO and B-rGO, suggesting oxygen atoms coming from B₂O₃ have been introduced into the graphene structure along with boron atoms. In the high resolution spectrum, the C1s peak of B-rGO (**Figure 1g**) was fitted to four components. The strong peaks located at 284.8 eV and 285.6 eV is assigned to sp² and sp³-bonded carbon atoms, respectively. The peak around 287 eV shows the existence of carbon-oxygen components, such as C-O, C=O, etc. A weak peak at 290.1 eV corresponds to the formation of C-O-B of B-rGO. The high resolution B1s spectrum of B-rGO was deconvoluted into three peaks located at 190.9 eV, 192.1 eV, and 193.0 eV, assigned to the BC₃, BC₂O or BCO₂, and BO₃, respectively (**Figure 1h**). The BC₃ component demonstrates that the boron atoms have been doped into carbon lattices. And the distinguishing peaks at 192.1 eV and 193.0 eV indicate boron atoms in form of oxide are bonded to carbon, verifying that the boron oxide are crosslinked with graphene sheets in B-rGO matrix.²⁷⁻²⁸ In order to further probe the boron-associated chemical bond, the B-rGO and pure B₂O₃ were investigated by the technique of EELS. In **Figure 2e and 2g**, compared with rGO, B K-edge is observed around an energy-loss of 193 eV in the spectrum of B-rGO layers.²⁹ Besides, carbon edge and oxygen edges are also found in the spectrum. As shown in **Figure 2d, 2f and 2h**, compared with pure B₂O₃, extra peaks of the δ*B-C and the π*B-C corresponding to B-C bonding are found, indicating boron atoms doped in the lattice

of graphene, and the existence of boron oxide in the rGO materials,³⁰⁻³¹ which are consistent with the XPS results. We also carried out EPR measurement, and found different signal profiles of rGO and B-rGO, further confirming the boron atoms were incorporated into the carbon lattices (**Figure S1** in the supporting information).

To investigate the discharge capacity of B-rGO in Li-O₂ batteries, voltage profiles of the initial discharge to 2.2 V (deep discharge) under different discharge rates were recorded, and compared with that of rGO, shown in **Figure 3a, 3b, 3c and 3d**. At the current density of 100 mA/g, B-rGO achieves an outstanding discharge capacity of ~ 18,000 mAh/g, while rGO reaches the discharge capacity of ~ 10,000 mAh/g, indicating that the B-rGO greatly increases energy density. When the current increases to 500 mA/g, the discharge capacity and voltage profile of B-rGO barely change, but the discharge capacity of rGO decreases much with a reduced ORR plateau. At the very high discharge rates (1000 mA/g and 2000 mA/g), the discharge capacities of both samples decrease. However, the capacity retention of B-rGO is far better than the one of rGO, and the ORR polarization of rGO becomes more intense than B-rGO. This result provides strong evidence that B-rGO greatly facilitates the ORR, which not only contributes to a larger discharge capacity but also is beneficial for a better rate capability.

To probe into the role of B-rGO in the improvement in ORR, we characterized and analyzed the discharge products harvested after the first discharge from the experiments above using the techniques of SEM (**Figure 3**), HR-XRD (**Figure 4**) and Raman spectroscopy (**Figure 5**). At the discharge current density of 100 mA/g, the

SEM images (**Figure 3a_i** and **3a_{ii}**) reveal that the morphology of the discharge products of both samples are toroid-like shape, which can be seen more clearly in the expanded images in the insets of **Figure 3a_i** and **3a_{ii}**. The size of the toroidal products of B-rGO ($\sim 2\ \mu\text{m}$) is apparently smaller than the ones of rGO ($\sim 4\ \mu\text{m}$), even though the discharge capacity of B-rGO is almost twice that of rGO. With the discharge rates increasing, the discharge products of both samples become smaller and smaller (**Figure 3b_i** and **3b_{ii}**). When the current density is beyond 500 mA/g, the discharge products from rGO change into a film-like shape (**Figure 3c_{ii}** and **3d_{ii}**). In contrast, the discharge products of B-rGO still maintain the toroidal shape (**Figure 3c_i** and **3d_i**). When the current density is increased to an extremely large value (2000 mA/g), the film-like discharge products begin to emerge in the B-rGO system (**Figure 3d_i**). The SEM results suggest B-rGO as the supporting material favors the maintaining of the toroidal shape than rGO does during high rates. The HR-XRD results of the discharged B-rGO samples confirm the formation of crystalline Li_2O_2 in all cases,^{11, 32} and the diffraction intensity of Li_2O_2 is subtly reduced with current increasing, as shown in **Figure 4**. However, the characteristic peaks of Li_2O_2 are clearly observed only from the rGO cathode discharged at the low current of 100 mA/g (**Figure 4a**). At the current density of 500 mA/g, the peaks of Li_2O_2 from rGO become very weak (**Figure 4b**). When the current increases beyond 500 mA/g, no peaks of Li_2O_2 are detected from the discharge electrodes by XRD (**4c** and **4d**). In Raman spectrum of the discharged electrodes of B-rGO, the peaks at $1123\ \text{cm}^{-1}$ and $1505\ \text{cm}^{-1}$ corresponding to LiO_2 -like and LiO_2 -graphite carbon surface components are

detected,³³⁻³⁴ regardless of the discharge rate, as shown in **Figure 5**. Nevertheless, only the weak peak associated to Li_2O_2 is observed from the discharged rGO electrodes at the low rate (**Figure 5a and 5b**), and the peak disappears when the current is beyond 500 mA/g (**Figure 5c and 5d**). Based on the analysis and discussions above, the discharge products from B-rGO are more likely to be toroids composed of LiO_2 -like and Li_2O_2 components in a wide range of discharge rates from 100 mA/g to 2000 mA/g. In contrast, the discharge products of rGO are mainly composed of toroidal Li_2O_2 at low currents, which change to poorly crystallized film-like $\text{Li}_{2-x}\text{O}_2$ at large currents.

Admittedly, FTIR spectra show the presence of some side reaction products on the discharged cathodes, including Li_2CO_3 (the bands at $\sim 850, 1450\text{ cm}^{-1}$) and $\text{CH}_3\text{CO}_2\text{Li}$ (the peaks located at around $650, 1650\text{ cm}^{-1}$), which are two typical by-products in the aprotic Li- O_2 system^{9, 35-36}, as shown in **Figure 6a and 6b**. In particular, the formation of Li_2CO_3 on the B-rGO cathode is less than the rGO one, especially at high discharge rates, and the amount of $\text{CH}_3\text{CO}_2\text{Li}$ is very small on both sample cathodes. Meanwhile XPS characterization of the fully discharged and charged cathodes at 100 mA/g was performed to obtain more information on the discharge products, as displayed in **Figure 7**. Both Li_2O_2 and Li_2CO_3 could be found in the Li 1s spectra of discharged rGO and B-rGO cathodes^{15, 35, 37-38}, while B-rGO cathode contains a greater proportion of Li_2O_2 to Li_2CO_3 than rGO, which is in agreement with FTIR results. Upon charge, only the peak of Li_2CO_3 could be deconvoluted from the Li 1s spectra, suggesting the complete decomposition of Li_2O_2 . So it may be concluded that

the B-rGO matrix could alleviate side reactions to some extent, compared with rGO. And the by-products, which are hard to avoid, present only in a very limited amount.

The results we got are strongly consistent with the through-solution growth mechanism proposed in other research.^{33, 39-43} Combining our results with the through-solution growth mechanism, we come up with a proper explanation for the phenomenon in this experiment, as illustrated in the **Figure 8a**. During discharge, the oxygen is reduced to O_2^- at the active sites on the surface of the matrix, and then O_2^- rapidly reacts with Li^+ to form LiO_2 . At low current rates, the formed LiO_2 dissolves into the electrolyte and transforms to Li_2O_2 through chemical disproportion, which begins to nucleate on the active sites of the electrode surface when the electrolyte is saturated. The precipitated Li_2O_2 serves as preferred nucleation sites for the continuous nucleation and crystallization of Li_2O_2 , resulting in the formation of the toroidal particles as observed in the SEM images. The presence of LiO_2 -like component in the discharge product of B-rGO may be due to more active sites from the defects and oxygen-based functional groups in the system as compared with rGO (**Figure 8b and 8c**). Consequently, the formation of LiO_2 on the surface of B-rGO may occur at a faster rate than its disproportion, leading to LiO_2 's deposition onto the surface of Li_2O_2 clusters. Meanwhile B-rGO provides more active sites and a larger specific surface area for Li_2O_2 nucleation and growth than rGO does, resulting in the formation of more but smaller toroids in the B-rGO system. Nazar⁴¹ has reported that the generation of solvated LiO_2 is controlled by the current rate. As the current is increased, the number of toroids increases and their relative size decreases, which is

in line with our results. At very high current densities, the poorly crystalized $\text{Li}_{x-2}\text{O}_2$ with a film morphology emerges. It may be due to the disproportionation of LiO_2 via surface migration, but there is still lack of strong evidence to support it. The peroxide film fully covers the surface of the substrate and is easy to completely curtail the discharge process due to its low electrical conductivity.⁴⁴ Owing to the 3D porous structure of B-rGO, its geometric surface area is far larger than rGO. As a result, at the same current the actual current density of B-rGO based on the geometric surface area is much lower, which is the reason that the B-rGO matrix, compared with the rGO matrix, favors maintaining the toroidal Li_2O_2 at increased currents.

The electrochemical performance of B-rGO and rGO as cathode was also investigated in $\text{Li}-\text{O}_2$ cells under capacity-controlled conditions. The first discharge-charge profiles of both samples were recorded at different rates with the fixed capacity of 1000 mAh/g, as shown in **Figure 9**. For the discharge stage, no obvious difference is found in the discharge plateaus (~ 2.75 V) between B-rGO and rGO at the low current density. As the current increases, the ORR overpotential of rGO becomes larger than that of B-rGO. This may be attributed to the presence of more defects or functional groups on the surface or edge of the B-rGO layers to activate the ORR process.^{16, 45} For the OER process, B-rGO really achieves a better rate performance with a reduced initial charge voltage, compared with rGO. At the large current density of 2000 mA/g, the charge capacity of rGO is only 180 mAh/g, while B-rGO could totally charge back. Three factors contribute to the excellent rate capability of B-rGO material. First, due to the 3D hierarchical structure, the B-rGO

material could facilitate the mass transfer process, and provide more active regions for reactions. Second, the toroidal discharge products from B-rGO are easier to decompose than the ones from rGO, which is attributed to their smaller size and LiO_2 -like component. Third, the most important of all, for Boron-doped rGO, the vacant orbital of B atoms extract the electrons from the graphite carbon π system through conjugation effect, leading to the electrons being greatly activated.⁴⁶⁻⁴⁷ Owing to the p-type behavior, B-rGO can promote charge transfer from Li_2O_2 to the substrate, decreasing energy barriers of rate determining steps (as illustrated in **Figure 8f**),²⁵ which is beneficial for fast charge under large current densities. As shown in Figure S4, the cycling performance of B-rGO is also superior to that of rGO. From the above discussions it can be seen that B-rGO as substrate is favorable for improving charge current density and rate capabilities of Li–O₂ batteries.

We also performed density functional theory (DFT) calculations to elucidate the role of boron-doped structure in improving the rate performance. The graphene layer (containing 54 carbon atoms, and hydrogen atoms are added to the edging to maintain electrical neutrality) and Li_5O_6 cluster (determined as the most suitable unit in recent report²⁶) are adopted as the models. The detailed information of the calculation is given in the Supporting Information. As shown in **Figure 8d and 8e**, optimized geometries of the Li_5O_6 cluster with Li-sited adsorption structure adsorbed on the surface of B-rGO and rGO are presented, and corresponding binding energy values are listed under the models. It is noteworthy that the binding energy of B-rGO (0.04006 Ha) is higher than the pristine one (0.02014 Ha), indicating that B-rGO

exhibits much stronger interaction with the Li_5O_6 cluster. Owing to the stronger adsorption for Li_2O_2 , B-rGO could more effectively activate Li-O bonds to decompose Li_2O_2 than rGO does during charge, resulting in a lower overpotential and better rate capability.

4. Conclusion

To summarize, the 3D porous B-rGO material with a hierarchical structure have been prepared by a facile freeze-drying method. In this design, the boric oxide produced from boric acid serves as the cross-linking agent to maintain the 3D porous architecture, facilitating the rapid diffusion of electrolyte and oxygen within the electrode. Owing to the plenty of functional groups and defects in the matrix, the B-rGO as cathode in $\text{Li}-\text{O}_2$ batteries achieves an extremely large discharge capacity and high discharge voltage plateau. Importantly, the boron atoms have been successfully incorporated into carbon lattices, leading to the electrons in the π system being greatly activated. Owing to the p-type behavior, B-rGO promotes charge transfer from Li_2O_2 to the substrate, which decreases energy barriers of rate determining steps. Meanwhile, B-rGO exhibits a much stronger interaction with the Li_5O_6 cluster, and thus B-rGO activates Li-O bonds to decompose Li_2O_2 more effectively than rGO does during charge. These are the two main contributors to the good rate capability and cycling life of B-rGO. With the advantages above, B-rGO could be an ideal substrate for $\text{Li}-\text{O}_2$ batteries. In future, we could add catalysts into the B-rGO matrix to further lower the overpotential and enhance the cycling performance.

ASSOCIATED CONTENT

Supporting Information

EPR spectra of B-rGO and rGO; HR-XRD of pristine B-rGO and rGO electrodes; Raman spectra of pristine B-rGO and rGO electrodes; The discharge and charge capacity versus cycle number of Li-O₂ batteries with rGO and B-rGO as cathode at different current densities; The first discharge-charge profiles of Li-O₂ batteries with B₂O₃-rGO as cathode at different current densities. This material is available free of charge via the Internet at <http://pubs.acs.org>.

AUTHOR INFORMATION

Corresponding Author

E-mail address: chenrj@bit.edu.cn (R.Chen), junlu@anl.gov (J.Lu)

Author Contributions

F.W, Y. X contributed equally to this work.

Notes:

The authors declare no competing financial interest.

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B and N Isolate-Doped Graphitic Carbon Nanosheets from Nitrogen-Containing
Ion-Exchanged Resins for Enhanced Oxygen Reduction. *Sci. Rep.* **2014**, *4*, 5184.



Scheme 1. Schematic diagram for the synthetic process of the B-rGO material.

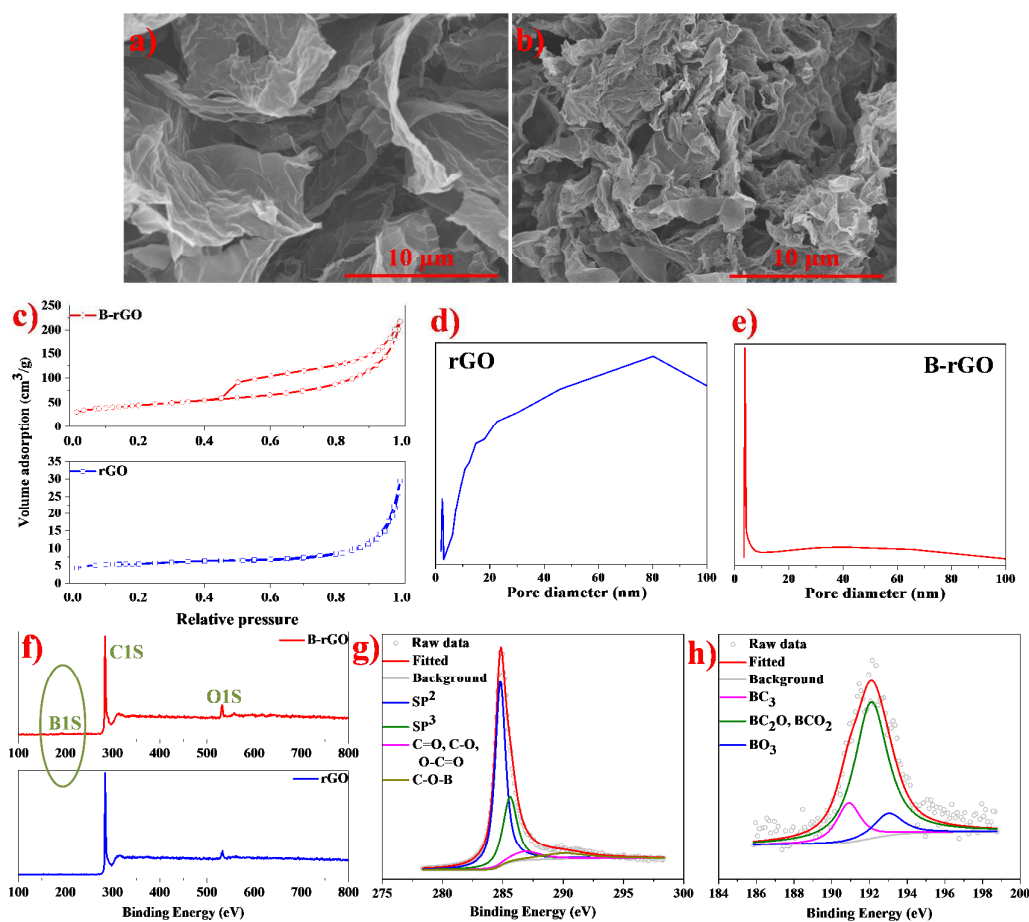


Figure 1. SEM images of the samples: a) rGO, b) B-rGO. (c) The nitrogen adsorption/desorption isotherms of rGO and B-rGO. The corresponding pore-size distribution of (d) rGO and (e) B-rGO. XPS spectra of B-rGO: (f) the sum, (g) C1s, (h) B1s.

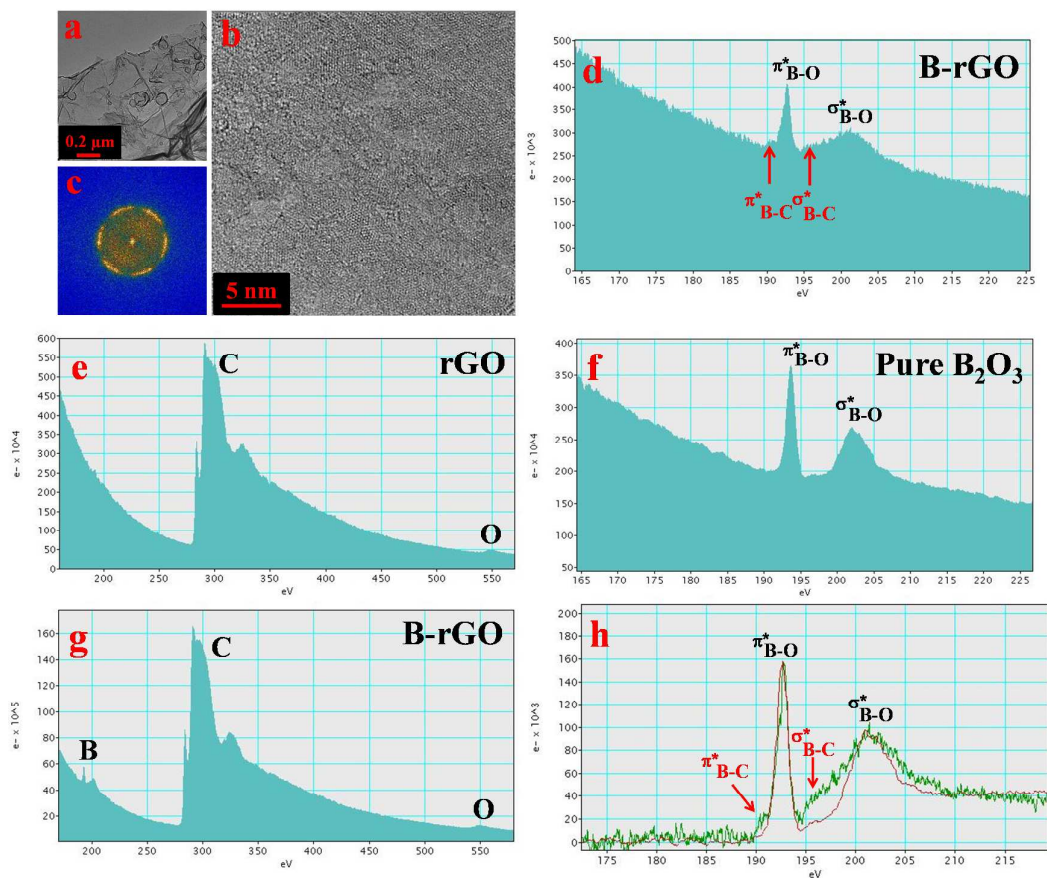


Figure 2. a) TEM image, b) High-resolution TEM image, c) SAED spectrum of B-rGO. EELS results for e) rGO, g) B-rGO. EELS spectra of B-K edges on d) B-rGO, f) pure B_2O_3 , h) corresponding combined figure.

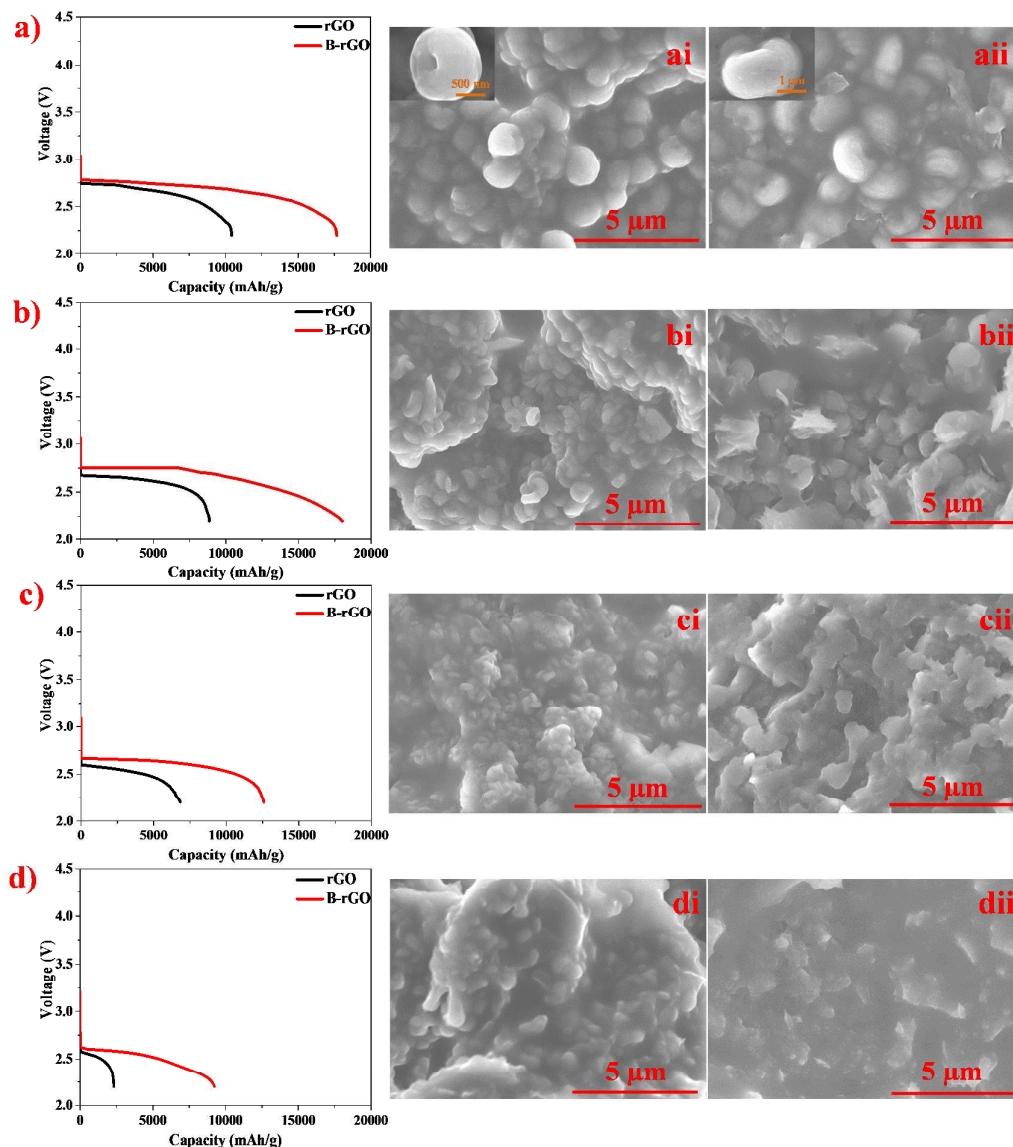


Figure 3. Discharge curves of Li-O₂ batteries with rGO and B-rGO as cathode to 2.2 V at current density of a) 100 mA/g, b) 500 mA/g, c) 1000 mA/g, d) 2000 mA/g. SEM images of discharge products harvested from a_i) B-rGO discharged at 100 mA/g to 2.2 V, a_{ii}) rGO discharged at 100 mA/g to 2.2 V, b_i) B-rGO discharged at 500 mA/g to 2.2 V, b_{ii}) rGO discharged at 500 mA/g to 2.2 V, c_i) B-rGO discharged at 1000 mA/g to 2.2 V, c_{ii}) rGO discharged at 1000 mA/g to 2.2 V, d_i) B-rGO discharged at 2000 mA/g to 2.2 V, d_{ii}) rGO discharged at 2000 mA/g to 2.2 V.

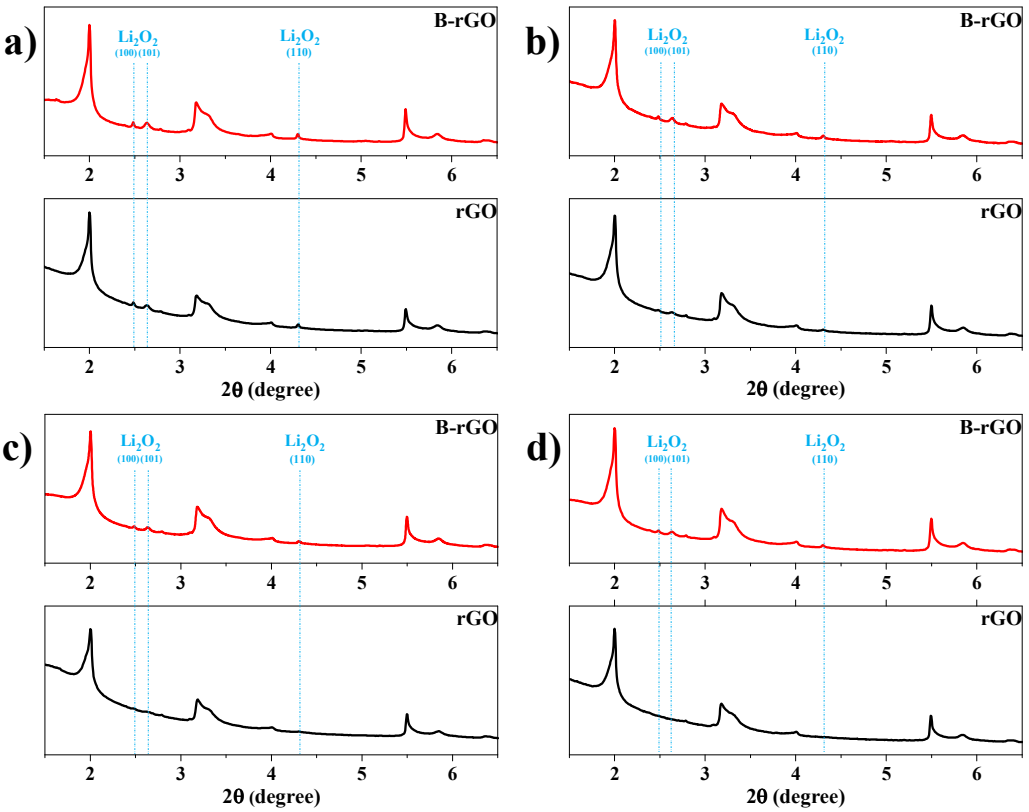


Figure 4. HR-XRD of B-rGO and rGO electrodes discharged to 2.2 V at current density of a) 100 mA/g, b) 500 mA/g, c) 1000 mA/g, d) 2000 mA/g.

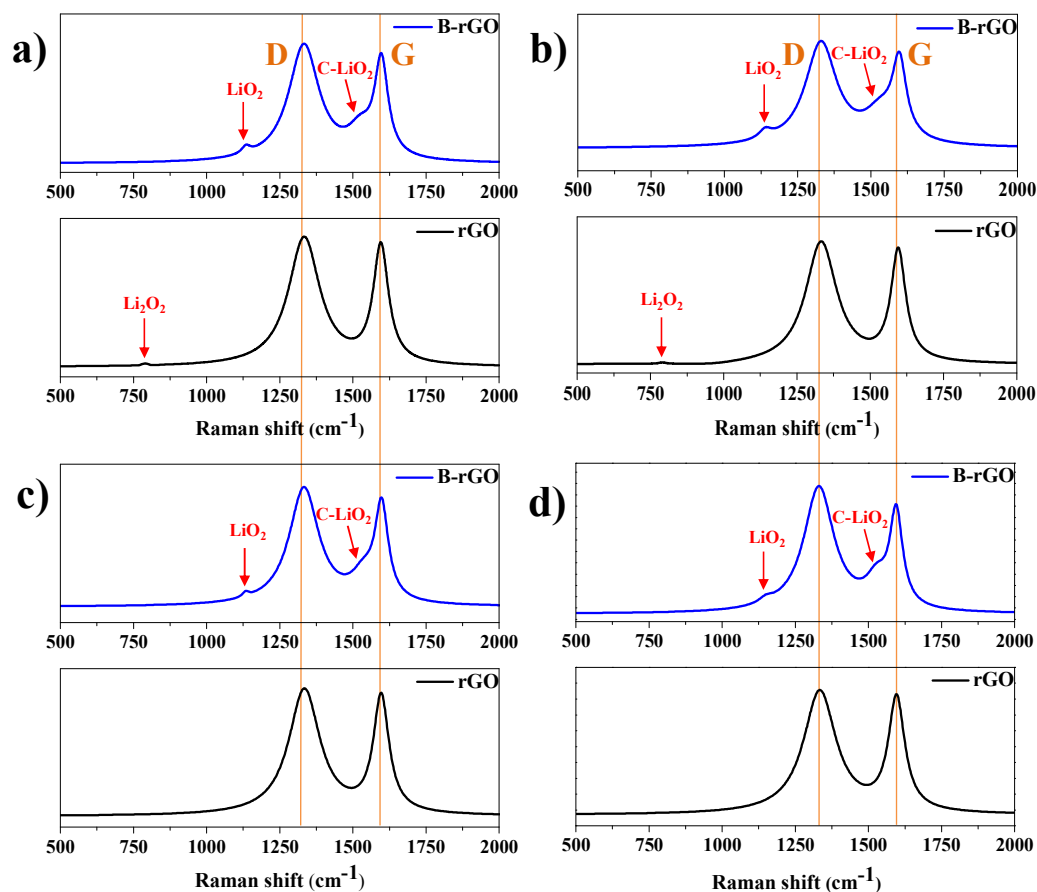


Figure 5. Raman spectra of B-rGO and rGO electrodes discharged to 2.2 V at current density of a) 100 mA/g, b) 500 mA/g, c) 1000 mA/g, d) 2000 mA/g.

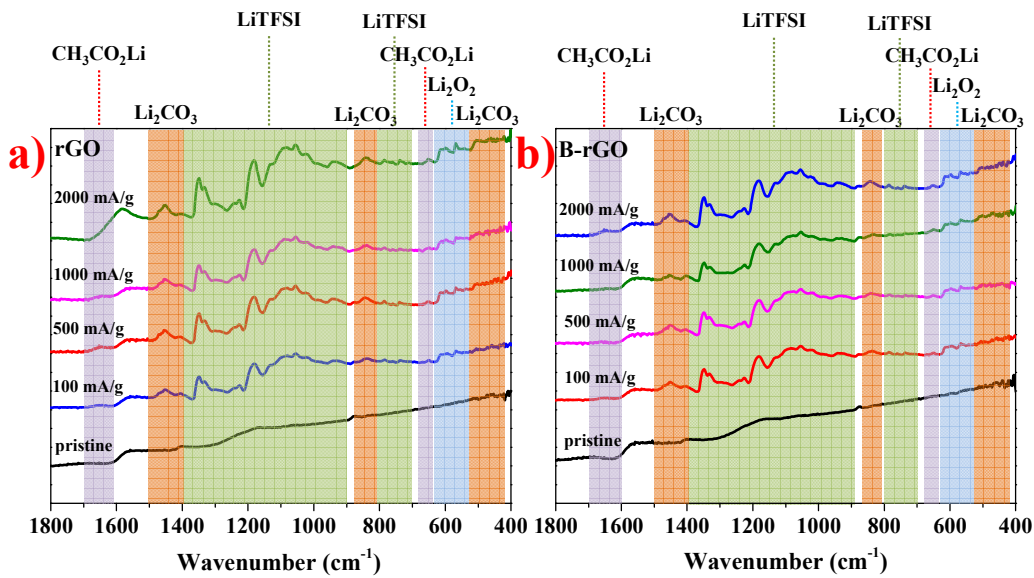


Figure 6. FTIR spectra of a) rGO and b) B-rGO electrodes discharged to 2.2 V at current density of 100 mA/g, 500 mA/g, 1000 mA/g, 2000 mA/g.

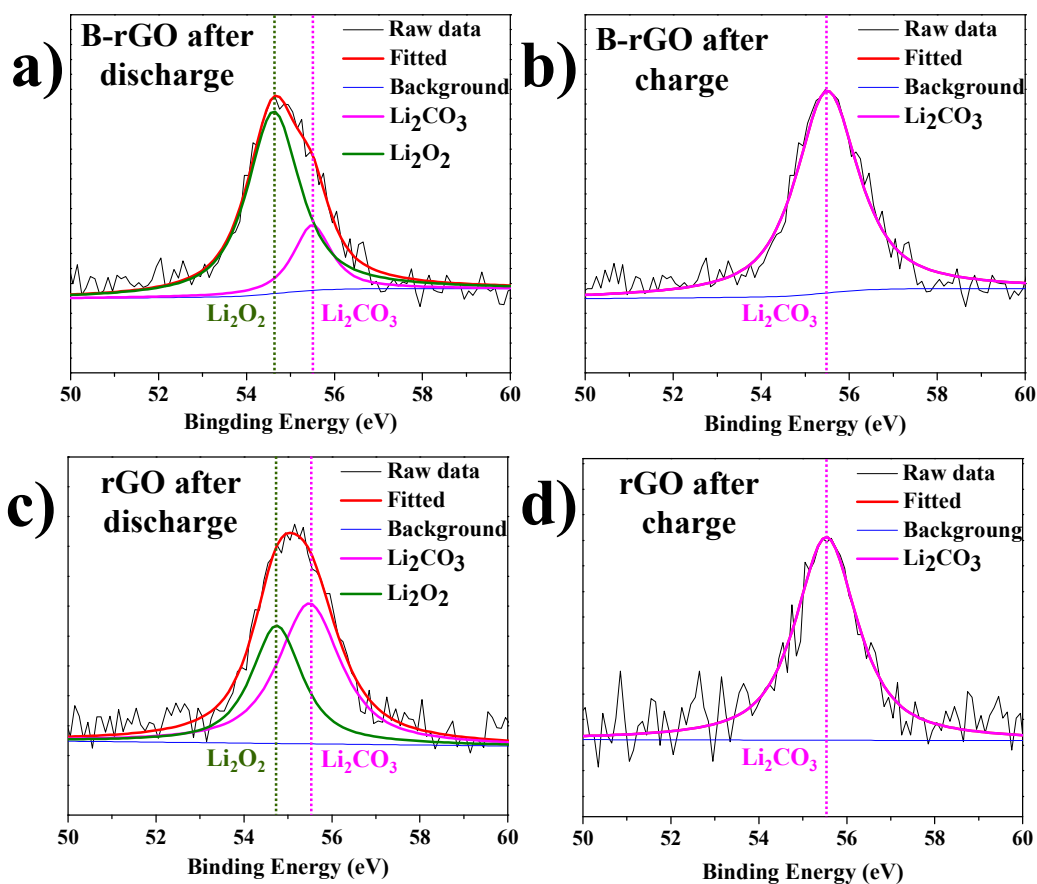


Figure 7. XPS spectra of fully discharged/charged cathode at current density of 100 mA/g: Li 1s spectra for (a) B-rGO cathode after discharge, (b) B-rGO cathode after recharge, (c) rGO cathode after discharge, (d) rGO cathode after recharge.

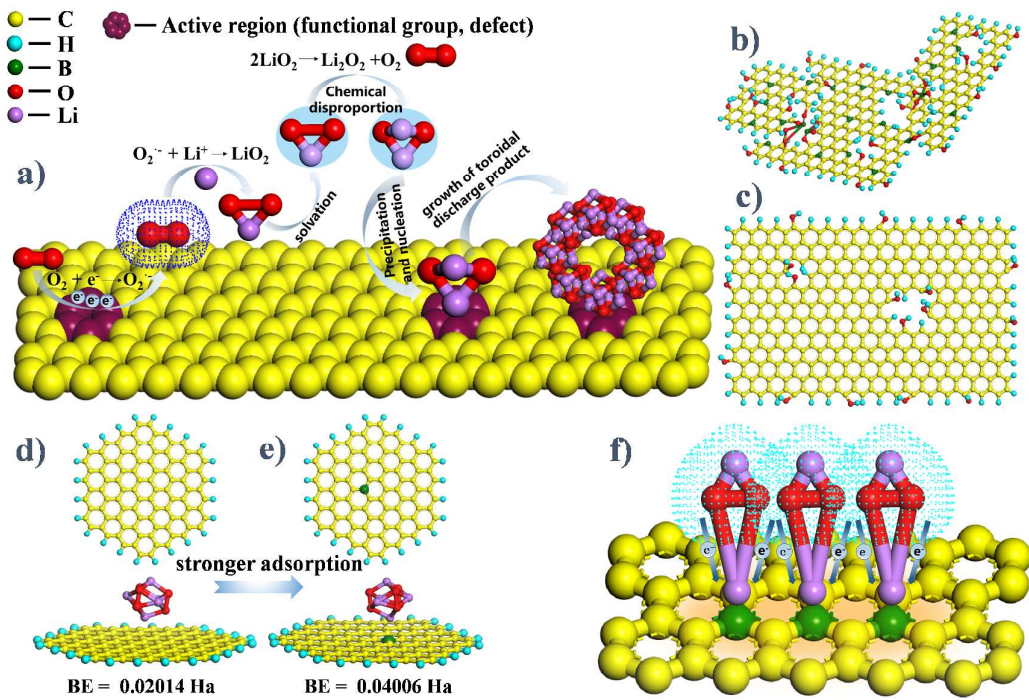


Figure 8. a) Schematic diagram of formation of the toroidal Li_2O_2 on the active region of the graphene layer. The ball-stick illustrations of b) B-rGO, c) rGO. Optimized geometries of the Li_5O_6 cluster and d) rGO, e) B-rGO. Corresponding binding energy values are also listed. f) Schematic picture of B-rGO as the substrate gaining electrons from the Li_2O_2 .

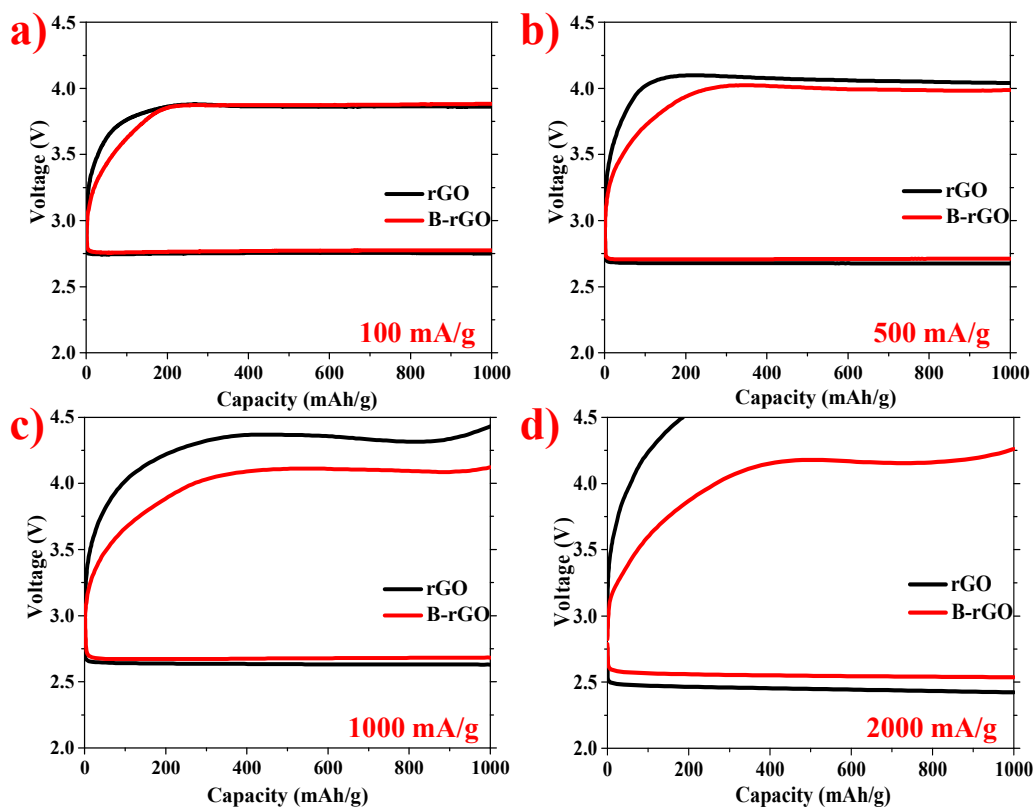


Figure 9. The first discharge-charge profiles of Li-O₂ batteries with B-rGO and rGO as cathodes by curtailing the capacity to 1000 mAh/g at current density of a) 100 mA/g, b) 500 mA/g, c) 1000 mA/g, d) 2000 mA/g.

Table of Contents Graphic

